

A Unified AI-Assisted Framework for Design Fairness Evaluation and Optimisation in Digital Interfaces

Venkatesh Akula

Student, Department of Computer Science & Engineering,
Indian Institute of Technology Kanpur
Kanpur 208016, Uttar Pradesh, India
Email: akulav22@iitk.ac.in

Sai Nikhil Sudhamandu

Student, Department of Computer Science & Engineering,
Indian Institute of Technology Kanpur
Kanpur 208016, Uttar Pradesh, India
Email: ssain22@iitk.ac.in

Bandaru Venkata Sritan

Student, Department of Electrical Engineering,
Indian Institute of Technology Kanpur
Kanpur 208016, Uttar Pradesh, India
Email: bandaruvs22@iitk.ac.in

Teja Sri Saladi

Student, Department of Electrical Engineering,
Indian Institute of Technology Kanpur
Kanpur 208016, Uttar Pradesh, India
Email: steja22@iitk.ac.in

Nayan Verma

Student, Department of Civil Engineering,
Indian Institute of Technology Kanpur
Kanpur 208016, Uttar Pradesh, India
Email: nayanv22@iitk.ac.in

Amar Kumar Behera

Associate Professor, Department of Design
Indian Institute of Technology Kanpur
Kanpur 208016, Uttar Pradesh, India
Email: amarkb@iitk.ac.in

Objectives

This research investigates how AI-assisted evaluation can support holistic design fairness in digital interfaces. The study addresses three questions:

- (1) Can accessibility, usability, and ethical design concerns be evaluated within a unified framework rather than through isolated tools?
- (2) Can agent-based simulation of user interaction reveal functional fairness issues missed by static analysis?
- (3) Can predictive remediation help designers reason about trade-offs between competing design principles?

Motivation

Digital interfaces are increasingly evaluated using fragmented tools—automated accessibility checkers, manual dark-pattern audits, and isolated visual tests—requiring designers to reconcile conflicting signals without systemic guidance (Zhang et al. 2023). Prior work in accessibility evaluation and ethical interface analysis highlights this fragmentation but offers limited support for understanding trade-offs across principles (Moon & Ahn 2025, Nakao et al. 2022). This reveals a gap in design research for integrated, decision-oriented frameworks that assist designers in navigating fairness as a multidimensional and interdependent design problem.

Methodology

A unified AI-assisted framework grounded in two design paradigms was developed: agentic auditing and predictive remediation. The system simulates representative user interactions, such as keyboard-only navigation and screen-reader usage, to identify functional accessibility and usability issues beyond DOM-based analysis. Detected issues are analysed alongside perceptual and ethical indicators. Generative models then propose minimal, actionable design and code revisions. Each proposed intervention is evaluated using a composite Design Fairness Score that hierarchically integrates technical, perceptual, and ethical signals to support informed decision-making.

Results and Conclusions

System-level evaluation demonstrates that agentic auditing uncovers a range of accessibility and ethical issues than static checks alone. Predictive remediation enables designers to compare potential fixes and understand trade-offs between competing design principles. The framework reframes interface auditing from a passive diagnostic task into proactive design decision support. This work contributes a scalable, human-centred approach for integrating fairness considerations into digital design practice.

References

- Zhang, J., Shu, Y., & Yu, H., 2023. Fairness in Design: A Framework for Facilitating Ethical Artificial Intelligence Designs. *Int. J. Crowd Sci.*, 7, pp. 32-39.
<https://doi.org/10.26599/ijcs.2022.9100033>
- Moon, D., & Ahn, S., 2025. Metrics and Algorithms for Identifying and Mitigating Bias in AI Design: A Counterfactual Fairness Approach. *IEEE Access*, 13, pp. 59118-59129.
<https://doi.org/10.1109/access.2025.3556082>.
- Nakao, Y., Strappelli, L., Stumpf, S., Naseer, A., Regoli, D., & Del Gamba, G. (2022). Towards Responsible AI: A Design Space Exploration of Human-Centered Artificial Intelligence User Interfaces to Investigate Fairness. *International Journal of Human-Computer Interaction*, 39, 1762 - 1788. <https://doi.org/10.1080/10447318.2022.2067936>.